DINITH SHEHAN JIFFRY

C++ Software Engineer | Financial Infrastructure & Exchange Systems

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PROFESSIONAL SUMMARY

Software Engineer with 5+ years working for the London Stock Exchange Group (LSEG) developing back-end financial infrastructure and clearing systems. Expertise in C++11, Linux system development, Apache Kafka connectivity, and low-level debugging using GDB and Valgrind. Familiar with conducting sprint ceremonies as a Scrum Master and developing developer tools to automate environment setup.

TECHNICAL SKILLS

- **Languages:** C++, Python, Java, SQL (PostgreSQL, Oracle).
- **Systems & Infrastructure:** Multithreading, Memory Management, Distributed Systems, Unix/Linux, Apache Kafka.
- **Debugging Tools:** GDB, Valgrind, Google Benchmark, Coverity, Git, Bash.
- **Domain:** Central Counterparty (CCP) Clearing, Market Infrastructure, Automated Testing, Agile/Scrum.

PROFESSIONAL EXPERIENCE

London Stock Exchange Group | Senior Software Engineer (CCP Team) | Mar 2024 – Present

- **A5X Core Platform Architecture:** Developed core C++ modules for exchange connectivity, including trade-leg correlation messaging pipelines, real-time MTM obligation enrichment, posting validations, and collateral response processing capabilities.
- **Data Structure and Latency Improvements:** Created in-memory caching of account linking references and reference data loaders, replacing high-latency database queries with fast in-memory lookup operations to remove I/O bottlenecks during trade volume peaks.
- **Kafka Messaging Library:** Developed a thread-safe and TLS encrypted C++ messaging library for optimized CSV and JSON array serialization/deserialization, minimizing dynamic memory allocations and locks during peak message throughput.
- **System Stability and Archival:** Added transactional data archiving and EOD data purging capabilities to control memory footprint and avoid database "blips" and "blobs" in case of heavy settlement windows.
- **Release Automation of AI:** Developed a release automation system that automatically manages the upgrade process of the framework in 8 C++11 project repos (4 core dependencies and 4 solution code bases).
- **Developer Tooling and Environments:** Automated topic, partitions, and mock server creation using Bash scripts and Kafka utilities, as well as an SQL stored procedure for automated system gateway creation.

- **Debugging and Code Quality:** Debugged and addressed edge-case memory leaks with GDB and Valgrind in multi-threaded applications.
- **Quality Governance & Defect Triage:** Transitioned Coverity static analysis from being used as a post hoc compliance check to a requirement before release; rigorously triaged scan results, eliminating false positives, and fixed critical memory and logic defects such that number of open issues is kept below thresholds.

London Stock Exchange Group | **Software Engineer (CCP Team)** | **Oct 2021 – Mar 2024**

- **Qatar Exchange Migration:** Programmed C++ back-end functionality and trading systems for Qatar Exchange migration to meet functional requirements for clearing processes.
- **European Clearing Services:** Responsible for maintenance of software for RCP/RCL (CCP in Paris and London) clearing systems and performing bug fixes.
- **Testing Framework:** Created unit tests, integration tests, and regression tests for the system in C++, taking care of edge cases under heavy load.
- **Scrum Master:** Acting as Scrum Master for CCP Development Team for one year and organizing daily stand-ups, planning and retrospectives

London Stock Exchange Group | **Software Engineer Intern** | **Feb 2019 – July 2019**

- **Benchmarking:** Measured STL container performance through Google Benchmark for the selection of containers used in the core engine.
- **Code Refactoring:** Refactored old C++ code bases by getting rid of warnings in the compiler.
- **SQL Database Migration:** Validated and migrated data sets from Oracle SQL to PostgreSQL.

EDUCATION

Sri Lanka Institute of Information Technology (SLIIT) | **Jan 2017 – August 2021**

- **BSc (Hons) in Information Technology (Software Engineering)**
- **GPA: 3.29/4.00**
- **Coursework:** Data Structures & Algorithms, Distributed Systems, Parallel Computing, OS Administration, Secure Software Development.

SYSTEMS PROJECTS

- **Turtlebot Maze Navigation (C++/ROS):** Developed a path finding algorithm utilizing ROS that enables the robot to navigate autonomously.
- **Fire Detection System (Java/API):** Constructed a multi-instance safety system capable of recovering from independent service failure.
- **Resource Security Access Control (C++/Java):** Developed an access control system with Apache Felix API for modular resource access.

CERTIFICATIONS & ACHIEVEMENTS

- **IELTS Academic (2021): Overall Band Score: 9.0** (Listening: 9.0, Reading: 8.0, Speaking: 8.0, Writing: 7.0).
- **Professional Development:** Certificates in Public Speaking and Personal/Professional Development (American College in Higher Education).